

CHAPTER 4: DYNAMIC ROUTING PROTOCOLS

Dynamic Routing is the process where routers automatically discover networks, build routing tables, and adapt to changes in the network topology without manual intervention from a network administrator.

THE THREE MAIN PROTOCOLS

1. RIP (ROUTING INFORMATION PROTOCOL)

RIP (Routing Information Protocol) is one of the oldest and simplest dynamic routing protocols. It is categorized as a **Distance-Vector** protocol, which means it makes routing decisions based on how far away a network is and which direction to send the data.

- **Distance-Vector (The "Rumor" Method)** – Routers only know two things: the **Distance** (how far) and the **Vector** (which direction/interface). They don't have a full map of the network; they only know what their neighbors tell them.

The most important thing to know about RIP is its **Metric**. A metric is the "score" a router uses to determine the best path.

- **Hop Count:** RIP simply counts the number of routers (hops) between the source and the destination.
- **The Winner:** If there are two paths to a destination, RIP will always choose the path with the **fewest number of routers**, even if that path is slower.

How RIP "Talks" (Updates)

RIP routers are very social, but not very efficient.

- **Full Updates:** Every **30 seconds**, a RIP router sends its entire routing table to its neighbors.
- **The "Rumor" Method:** A router only knows what its neighbor tells it. It doesn't have a map of the whole network; it just trusts its neighbor's directions.

RIP Versions

There are two main versions you will encounter in your labs:

1. **RIPv1:** The original version. It does not support **Subnetting (VLSM)**. It assumes all networks are standard Class A, B, or C.
2. **RIPv2:** The modern version. It supports **VLSM** (Variable Length Subnet Masking) and sends updates via **Multicast** (sending only to other routers), making it much more efficient.

2. OSPF (OPEN SHORTEST PATH FIRST)

OSPF (Open Shortest Path First) is a powerful, "intelligent" dynamic routing protocol used to connect large networks. It is categorized as a **Link-State** protocol, which is more advanced than the "Distance-Vector" approach used by RIP.

- **Link-State (The "Map" Method)** – Every router creates a complete, identical map of the entire network topology. They don't trade "directions"; they trade "status updates" about every link.

How OSPF Works: The "Link-State" Method

OSPF routers don't trade directions; they trade **Link-State Advertisements (LSAs)**.

- **The Map (LSDB):** Every OSPF router keeps a "Link-State Database." This is an identical copy of the entire network map.
- **The Math (Dijkstra's Algorithm):** Once the router has the map, it uses the **Shortest Path First (SPF)** algorithm to calculate the absolute fastest route to every destination.
- **The Metric (Cost):** OSPF doesn't care how many routers are in the way (hops). It cares about **Speed**. It assigns a "Cost" to every link based on bandwidth.
 - *Formula:* $\text{Cost} = 100,000,000 / \text{Bandwidth in bps}$
 - A 1Gbps link has a much lower "Cost" than a 10Mbps link. OSPF always picks the path with the lowest total cost.

To keep the "map" from getting too large and confusing for the router's brain (CPU), OSPF divides networks into **Areas**.

- **Area 0 (The Backbone):** This is the core area. All other areas must connect to Area 0.

3. EIGRP (ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL)

EIGRP (Enhanced Interior Gateway Routing Protocol) is a Cisco-proprietary routing protocol (though now an open standard) that is often called a **"Hybrid"** protocol. It combines the best parts of Distance-Vector (like RIP) and Link-State (like OSPF).

In the networking world, EIGRP is famous for being the **fastest** to recover if a cable is cut. While OSPF has to "re-calculate" the map, EIGRP usually has a backup plan already sitting in its memory.

Autonomous System (AS) Number

- Unlike OSPF where the Process ID (the 1) doesn't have to match, in EIGRP, the **AS Number MUST match on all routers**. If R1 is in AS 100 and R2 is in AS 200, they will ignore each other.

RTP (Reliable Transport Protocol)

- EIGRP is very organized. It uses its own delivery system (RTP) to ensure that when it sends a routing update, the other router sends back an "Acknowledgment" (ACK). If it doesn't get an ACK, it resends the data.

References:

- Odom, W. (2019). CCNA 200-301 official cert guide, volume 1. Cisco Press.